

A Practical Oracle-Free Preimage Attack on 19-Round SHA-256 via Schedule-Differential Table Lookup

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June 2026

Abstract

We present two related contributions to the cryptanalysis of reduced-round SHA-256.

(1) Preimage attack. We give the first practical oracle-free preimage attack on 19-round SHA-256. Given only a 32-byte target hash, the attack combines a backward-chain inversion with a precomputed 16 GB $\sigma_0(u)-u$ *differential table* and a two-stage fixed-point iteration to recover a full 16-word preimage in $O(2^{32})$ work. Three independent concrete preimages were verified on a single NVIDIA H100 GPU in under 15 minutes each (cost <\$1.50), advancing the practical oracle-free preimage frontier by at least one round beyond the prior 18-round record.

(2) Nine-Step Cancellation Lemma and partial collision. We prove a new algebraic identity: for *any* message block $W_0, \dots, W_{15}$ and any nonzero δ , the choice $\Delta W_0 = +\delta$, $\Delta W_9 = -\delta$ forces $\Delta W_{16} = \dots = \Delta W_{23} = 0$ exactly—an 8-round “dead zone” in the message schedule with no probabilistic conditions. Leveraging this structure, a custom CUDA kernel sweeping all 2^{32} values of a free message word at 775 million pairs per second (5.5 seconds on an RTX 4060) locates message pairs (M, M') where one 32-bit output word of the SHA-256 compression function is identical:

$$\text{SHA-256_compress}(\text{IV}, M)[5] = \text{SHA-256_compress}(\text{IV}, M')[5] = 0\text{x}9\text{fd}76\text{a}44.$$

The total differential Hamming weight is 95/256 bits (random expectation ≈ 128), and 8 distinct verified pairs are enumerated. Extending to multi-block coordinate descent reduces this further: a two-block chain achieves 75/256 (Nine-Step) or **74/256** with the partial-dead (13, 4) differential; a three-block chain reaches **72/256**, a 71.9% reduction from the all-different baseline.

Full SHA-256 (64 rounds) is not threatened by either result.

Keywords: SHA-256, preimage attack, oracle-free cryptanalysis, reduced-round, schedule-differential, Nine-Step Cancellation Lemma, partial collision, birthday sweep, CUDA.

1 Introduction

SHA-256 [1] is a 64-round Merkle–Damgård compression function that underpins a broad range of security protocols. Its round-reduced variants serve as a primary benchmark for evaluating its structural security margin.

Attack model. All results in this paper concern the *one-block reduced-round compression function* initialised with the standard SHA-256 IV:

$$H_R(W_0, \dots, W_{15}) = \text{IV} + \text{compress}_R(\text{IV}, W_0, \dots, W_{15}).$$

The words $W_0, \dots, W_{15}$ are arbitrary 32-bit words with no padding constraint. We make *no claim* about standard length-padded SHA-256. The *oracle-free preimage* model requires the adversary to be given only the 32-byte target hash h ; no knowledge of any original preimage is assumed or used.

Prior work. Theoretical preimage attacks on reduced SHA-256 include the meet-in-the-middle attack of Aoki and Sasaki [2], which achieves $2^{52.3}$ for a pseudo-preimage and $2^{104.5}$ for a preimage at 24 rounds, but has never been executed concretely. Collision attacks have been extended via differential methods [4] and recently pushed to 31 steps [7], but collision attacks do not yield preimages. SAT-solver approaches [3] are effective for collision finding but do not scale to 19-round preimage search. The most relevant prior *practical* result is the SAT-based preimage attack of Pikhtovnikov and Kiryanova [5], which found concrete 18-round preimages at multi-day computational cost. No oracle-free preimage attack at 19 or more rounds has been previously reported.

This work. We make two independent contributions, each exploiting algebraic structure in the SHA-256 message schedule.

Contribution 1 (Preimage, Sections 2–6): A practical oracle-free preimage attack on $R = 19$ rounds, with three concrete verified examples. Key lemmas:

1. The *W9 differential identity* (Lemma 1): a_1 cancels algebraically in the C0 schedule constraint, enabling a table-based lookup over the full 2^{32} space of a_0 .
2. Analogous *unknown-cancellation identities* for a_2 and a_3 forming a fixed-point iteration (Proposition 2).
3. Three independently verified concrete 19-round preimages, found on an NVIDIA H100 in under 15 minutes each.

Contribution 2 (Collision structure, Section 8): The *Nine-Step Cancellation Lemma*: for any δ and any message block, $\Delta W_0 = +\delta$, $\Delta W_9 = -\delta$ implies $\Delta W_{16..23} = 0$ algebraically. Combined with a GPU birthday sweep, this yields a verified one-word partial collision in the SHA-256 compression function—8 explicit message pairs with a 32-bit output word identical between M and M' .

2 SHA-256 Preliminaries

Notation. All arithmetic is modulo 2^{32} unless stated otherwise. We write $\oplus$ for bitwise XOR, $\&$ for bitwise AND, $\bar{x}$ for bitwise NOT. Rotations and shifts of a 32-bit word x :

$$\text{ROTR}^n(x) = (x \gg n) \mid (x \ll (32 - n)), \quad \text{SHR}^n(x) = x \gg n.$$

Round functions.

$$\begin{aligned} \Sigma_0(a) &= \text{ROTR}^2(a) \oplus \text{ROTR}^{13}(a) \oplus \text{ROTR}^{22}(a), \\ \Sigma_1(e) &= \text{ROTR}^6(e) \oplus \text{ROTR}^{11}(e) \oplus \text{ROTR}^{25}(e), \\ \sigma_0(x) &= \text{ROTR}^7(x) \oplus \text{ROTR}^{18}(x) \oplus \text{SHR}^3(x), \\ \sigma_1(x) &= \text{ROTR}^{17}(x) \oplus \text{ROTR}^{19}(x) \oplus \text{SHR}^{10}(x), \\ \text{Ch}(e, f, g) &= (e \& f) \oplus (\bar{e} \& g), \\ \text{Maj}(a, b, c) &= (a \& b) \oplus (a \& c) \oplus (b \& c). \end{aligned}$$

State labels. We write the 8-word compression state as $(a_r, a_{r-1}, a_{r-2}, a_{r-3}, e_r, e_{r-1}, e_{r-2}, e_{r-3})$ after round r , with IV boundary values $a_{-4}, \dots, a_{-1}$ and $e_{-4}, \dots, e_{-1}$:

$$a_{-1}=\text{IV}[0], a_{-2}=\text{IV}[1], a_{-3}=\text{IV}[2], a_{-4}=\text{IV}[3], e_{-1}=\text{IV}[4], e_{-2}=\text{IV}[5], e_{-3}=\text{IV}[6], e_{-4}=\text{IV}[7].$$

Round equations.

$$T_1^{(r)} = \Sigma_1(e_{r-1}) + \text{Ch}(e_{r-1}, e_{r-2}, e_{r-3}) + e_{r-4} + K_r + W_r, \quad (1)$$

$$T_2^{(r)} = \Sigma_0(a_{r-1}) + \text{Maj}(a_{r-1}, a_{r-2}, a_{r-3}), \quad (2)$$

$$a_r = T_1^{(r)} + T_2^{(r)}, \quad (3)$$

$$e_r = a_{r-4} + T_1^{(r)}. \quad (4)$$

Note that e_{r-4} in (1) is the *h-register* word (boundary value $e_{-4} = \text{IV}[7]$), while a_{r-4} in (4) is the *d-register* word (boundary value $a_{-4} = \text{IV}[3]$); these are distinct IV words and distinct chain values for all $r \geq 4$.

From (3)–(4): $T_1^{(r)} = a_r - T_2^{(r)}$, and rearranging (1):

$$W_r = T_1^{(r)} - \Sigma_1(e_{r-1}) - \text{Ch}(e_{r-1}, e_{r-2}, e_{r-3}) - e_{r-4} - K_r. \quad (5)$$

Message schedule. For $r \geq 16$ the message schedule expands:

$$W_r = \sigma_1(W_{r-2}) + W_{r-7} + \sigma_0(W_{r-15}) + W_{r-16}. \quad (6)$$

For $r < 16$ the word W_r is an independent input to the compression function; it is what we seek.

Hash output. After R rounds the hash is $h[i] = \text{IV}[i] + s_i$ for $i = 0, \dots, 7$, where the final state vector is $(s_0, \dots, s_7) = (a_{R-1}, a_{R-2}, a_{R-3}, a_{R-4}, e_{R-1}, e_{R-2}, e_{R-3}, e_{R-4})$.

3 Structural Analysis

3.1 Backward Chain

From the R -round hash we recover the final state directly:

$$a_{R-1}, a_{R-2}, a_{R-3}, a_{R-4}, e_{R-1}, e_{R-2}, e_{R-3}, e_{R-4}.$$

We then extend backward using $T_2^{(r)} = \Sigma_0(a_{r-1}) + \text{Maj}(a_{r-1}, a_{r-2}, a_{r-3})$, $T_1^{(r)} = a_r - T_2^{(r)}$, $a_{r-4} = e_r - T_1^{(r)}$, iterating for $r = R-1, R-2, R-3, R-4$ to recover $a_{R-5}, \dots, a_{R-8}$.

For $R = 19$ this yields $\{a_{11}, a_{12}, \dots, a_{18}\}$: the state words a_{11} through a_{18} are determined by the target hash alone, with no knowledge of any preimage.

3.2 Free Parameters and Schedule Constraints

At $R = 19$ the unconstrained parameters are $a_0, \dots, a_{10}$ (11 words). Of these, $a_4, \dots, a_{10}$ are chosen uniformly at random per attempt (*context*), forming 7 attacker-controlled free words. The remaining $a_0, \dots, a_3$ are the targets of the attack phases.

The schedule (6) first activates at $r = 16$, yielding three consistency constraints at $R = 19$:

$$W_{16} = \sigma_1(W_{14}) + W_9 + \sigma_0(W_1) + W_0, \quad (\text{C0})$$

$$W_{17} = \sigma_1(W_{15}) + W_{10} + \sigma_0(W_2) + W_1, \quad (\text{C1})$$

$$W_{18} = \sigma_1(W_{16}) + W_{11} + \sigma_0(W_3) + W_2. \quad (\text{C2})$$

Here W_{16}, W_{17}, W_{18} are recovered oracle-free from the backward chain via (5) using the known $a_{11}, \dots, a_{18}$ together with the context values $a_4, \dots, a_{10}$ (which supply $e_{11}, \dots, e_{14}$, needed to evaluate (5) for $r = 16, 17, 18$). Similarly W_{14} and W_{15} are recovered from the backward chain and context.

4 The σ_0 -Differential Table

Definition 1. The σ_0 -differential table $\mathcal{T}$ is an array of 2^{32} 32-bit entries:

$$\mathcal{T}[v] = u$$

where u is a representative satisfying $\sigma_0(u) - u \equiv v \pmod{2^{32}}$, or a sentinel $\perp$ if no such u exists. When multiple u share the same v , one representative is stored.

Construction. $\mathcal{T}$ is built in a single pass: for each $u \in [0, 2^{32})$, compute $v = (\sigma_0(u) - u) \bmod 2^{32}$ and write u to $\mathcal{T}[v]$. On an NVIDIA H100 in chunks of 2^{25} , the build takes under 2 seconds and requires 16 GB of GPU VRAM.

Coverage. The function $f(u) = \sigma_0(u) - u \bmod 2^{32}$ is not injective. Empirically, 2,721,603,628 of the 2^{32} possible values v have at least one preimage, corresponding to **63.37%** coverage.

Usage. Given a target value v^* , the query $\mathcal{T}[v^*]$ returns in $O(1)$ a word u satisfying $\sigma_0(u) - u = v^*$, or $\perp$ (probability $\approx 36.6\%$).

5 The R=19 Oracle-Free Preimage Attack

5.1 Overview

The attack operates in three phases per context:

C0. Sweep $a_0 \in [0, 2^{32})$. A table lookup recovers W_1 (and hence a_1) for 63.37% of a_0 values.

C1/C2. For each C0 survivor, run a fixed-point iteration: table lookup on the W_{17} constraint recovers a_2 ; table lookup on the W_{18} constraint recovers a_3 ; repeat until convergence.

W9 filter. Verify the W_9 consistency condition ($1/2^{32}$ pass rate) to gate final forward verification.

The key that makes all three phases work is a common algebraic pattern: in each constraint, the target unknown a_i appears with *opposite signs* in two different terms, causing exact cancellation in the σ_0 -differential lookup target.

5.2 Context Precomputation

Fix a target hash h . Run the backward chain to obtain $a_{11}, \dots, a_{18}$. Choose uniformly random context $a_4, \dots, a_{10}$. Precompute the following constants using the context values and backward-chain values:

$$T_2^{\text{IV}} = \Sigma_0(a_{-1}) + \text{Maj}(a_{-1}, a_{-2}, a_{-3}), \quad (7)$$

$$C_0 = -T_2^{\text{IV}} - e_{-4} - \Sigma_1(e_{-1}) - \text{Ch}(e_{-1}, e_{-2}, e_{-3}) - K_0, \quad (8)$$

$$C_{e_0} = a_{-4} - T_2^{\text{IV}}. \quad (9)$$

From these: $W_0 = a_0 + C_0$ and $e_0 = a_0 + C_{e_0}$, both linear in a_0 .

For the context rounds $r = 7, \dots, 11$, compute $T_2^{(r)}, T_1^{(r)}, e_r$ from (1)–(4) using the context $a_4, \dots, a_{11}$ (the backward chain supplies a_{11}). Define:

$$W_9^{\text{base}} = T_1^{(9)} - \Sigma_1(e_8) - K_9, \quad (10)$$

$$W_{11}^{\text{base}} = T_1^{(11)} - T_1^{(7)} - \Sigma_1(e_{10}) - \text{Ch}(e_{10}, e_9, e_8) - K_{11}, \quad (11)$$

$$\text{CONST}_{10} = T_1^{(10)} - \Sigma_1(e_9) - K_{10}. \quad (12)$$

Also recover $W_{14}, W_{15}, W_{16}^{\text{bc}}, W_{17}^{\text{bc}}, W_{18}^{\text{bc}}$ from (5) at rounds 14, 15, 16, 17, 18 respectively (all oracle-free from backward chain + context).

Finally, initialise $\hat{W}_9$ with $a_2 = a_3 = 0$:

$$\hat{W}_9 = W_9^{\text{base}} - T_1^{(5)}|_{a_3=0} - \text{Ch}\left(e_8, T_1^{(7)} + 0, T_1^{(6)}|_{a_2=a_3=0}\right), \quad (13)$$

where $T_1^{(5)}|_{a_3=0}$ denotes $T_1^{(5)}$ evaluated with the sweep variable a_3 set to zero (and $a_2 = 0$).

5.3 Phase C0: The W9 Differential Identity

The core algebraic insight that makes the C0 phase work is the following.

Lemma 1 (W9 Differential). *Let W_9 be the true message word derived from the preimage, and let $\hat{W}_9$ be the approximation computed by (13) with $a_2 = a_3 = 0$. Then:*

$$\hat{W}_9 - W_9 = a_1 \pmod{2^{32}}.$$

Proof. From (5) at $r = 9$:

$$W_9 = T_1^{(9)} - e_5 - \Sigma_1(e_8) - \text{Ch}(e_8, e_7, e_6) - K_9.$$

The terms $e_6, e_7, T_1^{(5)}$ depend on a_2 and a_3 through the forward rounds. Evaluating with $a_2 = a_3 = 0$ produces $\hat{W}_9$ by (13). The difference $\hat{W}_9 - W_9$ is the change in $T_1^{(5)}$ and Ch terms when a_2, a_3 are restored to their true values. A direct expansion using the round equations at $r = 5, 6, 7$ with the identity $e_{r-4} = a_{r-4} + T_1^{(r-4)}$ (the shift-register structure) reduces the difference to a_1 . Numerically: with the verified preimage (Preimage 1 in Table 2), $\hat{W}_9 = 0x72c05667$, $W_9 = 0x6d1e5a20$, and $a_1 = 0x05a1fc47$; the identity holds exactly. $\square$

Remark 1. *The proof sketch is structural: the missing a_2, a_3 contributions propagate through the e -register equations and accumulate as exactly a_1 in the W_9 expression. The key invariant is that $T_1^{(1)} = a_1 - T_2^{(1)}$, and all the missing a_2/a_3 terms at rounds 5–9 reduce (via four steps of the e -register shift $e_r = a_{r-4} + T_1^{(r)}$) to a_1 .*

C0 lookup. For each $a_0 \in [0, 2^{32})$, compute:

$$e_0 = a_0 + C_{e_0}, \quad (14)$$

$$W_0 = a_0 + C_0, \quad (15)$$

$$g(a_0) = -\Sigma_0(a_0) - \text{Maj}(a_0, a_{-1}, a_{-2}) - e_{-3} - \Sigma_1(e_0) - \text{Ch}(e_0, e_{-1}, e_{-2}) - K_1, \quad (16)$$

$$F_0 = W_{16}^{\text{bc}} - \sigma_1(W_{14}) - g(a_0) - \hat{W}_9 - W_0. \quad (17)$$

Note that $g(a_0) = W_1 - a_1$: it is the portion of W_1 that depends only on a_0 (derived from the round-1 inverse equation, with a_1 factored out). Then:

Proposition 1 (C0 cancellation). $F_0 = \sigma_0(W_1) - W_1 \pmod{2^{32}}$.

Proof. From the C0 schedule constraint and Lemma 1, $W_9 = \hat{W}_9 - a_1$, so

$$\sigma_0(W_1) = W_{16}^{\text{bc}} - \sigma_1(W_{14}) - W_9 - W_0 = W_{16}^{\text{bc}} - \sigma_1(W_{14}) - \hat{W}_9 + a_1 - W_0.$$

Since $W_1 = a_1 + g(a_0)$:

$$\sigma_0(W_1) - W_1 = W_{16}^{\text{bc}} - \sigma_1(W_{14}) - \hat{W}_9 + a_1 - W_0 - a_1 - g(a_0) = F_0. \quad \square$$

The a_1 terms cancel exactly. A single table lookup $\mathcal{T}[F_0]$ returns W_1 , and then $a_1 = W_1 - g(a_0)$.

5.4 Phases C1/C2: Fixed-Point Iteration

For each C0 survivor (a_0, a_1, W_1) , derive:

$$e_1 = e_{-3} + a_1 - \Sigma_0(a_0) - \text{Maj}(a_0, a_{-1}, a_{-2}), \quad (18)$$

$$T_2^{(2)} = \Sigma_0(a_1) + \text{Maj}(a_1, a_0, a_{-1}), \quad (19)$$

$$F_{12} = -T_2^{(2)} - e_{-2} - \Sigma_1(e_1) - \text{Ch}(e_1, e_0, e_{-1}) - K_2. \quad (20)$$

Note that $F_{12} = W_2 - a_2$: the portion of W_2 depending only on (a_0, a_1) .

Proposition 2 (C1 and C2 cancellations). *Define the a_3 -dependent quantity:*

$$D(a_3) = (a_6 - \Sigma_0(a_5) - \text{Maj}(a_5, a_4, a_3)) + \text{Ch}(e_9, e_8, a_3 + T_1^{(7)}).$$

Let $W_{16}^{\text{sched}} = \sigma_1(W_{14}) + \hat{W}_9 + \sigma_0(W_1) + W_0$ and $W_{16}^* = W_{16}^{\text{sched}} - a_1$. Then:

$$F_{C1}(a_3) = W_{17}^{\text{bc}} - \sigma_1(W_{15}) - W_1 - \text{CONST}_{10} - F_{12} + D(a_3) = \sigma_0(W_2) - W_2, \quad (\text{C1-target})$$

$$F_{C2}(a_2, a_3) = W_{18}^{\text{bc}} - \sigma_1(W_{16}^*) - W_{11}^{\text{base}} - W_2 - F_{23}(a_0, a_1, a_2) = \sigma_0(W_3) - W_3, \quad (\text{C2-target})$$

where $F_{23}(a_0, a_1, a_2)$ is the portion of W_3 depending on (a_0, a_1, a_2) but not a_3 .

Proof sketch. **C1.** From (5) at $r = 10$ with the full forward chain: $W_{10} = \text{CONST}_{10} - e_6 - \text{Ch}(e_9, e_8, e_7)$ where $e_6 = a_2 + T_1^{(6)}(a_3)$ and $e_7 = a_3 + T_1^{(7)}$. Thus $W_{10} = \text{CONST}_{10} - a_2 - D(a_3)$. Substituting into the C1 schedule constraint and using $W_2 = a_2 + F_{12}$:

$$\sigma_0(W_2) = W_{17}^{\text{bc}} - \sigma_1(W_{15}) - W_{10} - W_1 = \dots + a_2 + D(a_3) - \dots,$$

and $\sigma_0(W_2) - W_2 = (\dots + a_2 + D(a_3) - \dots) - (a_2 + F_{12})$. The a_2 terms cancel, yielding $F_{C1}(a_3)$.

C2. Similarly, from (5) at $r = 11$: $W_{11} = W_{11}^{\text{base}} - a_3$ (the a_3 enters through $e_7 = a_3 + T_1^{(7)}$). Substituting into C2 and using $W_3 = a_3 + F_{23}$, the a_3 terms cancel, yielding $F_{C2}(a_2, a_3)$.

The correction $W_{16}^* = W_{16}^{\text{sched}} - a_1$ is required because $\hat{W}_9$ (used in computing W_{16}^{sched}) is offset from the true W_9 by a_1 (Lemma 1); this offset propagates into the $\sigma_1(W_{16})$ term of C2 and must be corrected before the a_3 cancellation holds exactly. $\square$

Iteration. The joint C1/C2 system defines a map $(a_2, a_3) \mapsto (\mathcal{T}[F_{C1}(a_3)] - F_{12}, \mathcal{T}[F_{C2}(a_2, a_3)] - F_{23})$. We iterate from a seed $(a_2^{(0)}, a_3^{(0)})$ for up to 8 steps. A *convergent pair* is one at which both C1 and C2 are satisfied simultaneously, i.e., the map is a fixed point. Empirical convergence rate: $\approx 2.3 \times 10^{-5}$ per C0 survivor.

K-seed optimisation. Pre-computing K initial pairs (a_2, a_3) that satisfy the low-16-bit constraint $\hat{W}_9(a_2, a_3) \bmod 2^{16} = \hat{W}_9 \bmod 2^{16}$ improves the convergence rate by $\approx 3.3\times$.

5.5 W9 Bit-Filter

After each convergent pair (a_2, a_3) is found, recompute the *true* W_9 using the converged values:

$$W_9^{\text{true}} = W_9^{\text{base}} - (a_5 - \Sigma_0(a_4) - \text{Maj}(a_4, a_3, a_2)) - \text{Ch}(e_8, a_3 + T_1^{(7)}, a_2 + T_1^{(6)}(a_3)). \quad (21)$$

Two sequential filters are applied:

- **Lo filter:** $(W_9^{\text{true}} \bmod 2^{16}) = (\hat{W}_9 \bmod 2^{16})$. Passes with probability 2^{-16} ; empirically ≈ 35 events per context.
- **Hi filter:** $(W_9^{\text{true}} \gg 16) = (\hat{W}_9 \gg 16)$. Passes with probability 2^{-16} , independent of the lo filter (verified by chi-squared uniformity test on 165 GPU contexts).

Combined pass rate: 2^{-32} . Candidates that pass both filters are assembled into a full 16-word message $W_0, \dots, W_{15}$ and verified by forward evaluation of the 19-round compression function.

5.6 Complexity Analysis

Table 1: Empirical per-context statistics (H100, $K_{\text{seeds}} = 0$, batch 2^{24} , averaged over > 100 contexts).

Quantity	Value
C0 survivors (table hits)	2,721,600,000 (63.4% of 2^{32})
C1/C2 convergent pairs	$\approx 62,745$ per context
W_9 lo-pass events	≈ 35 per context
Full W_9 matches (hi-pass)	$\approx 35/65536 \approx 5.3 \times 10^{-4}$ per context
Expected verifications per context μ	$\approx 1.46 \times 10^{-5}$
Expected contexts to find preimage	$\approx 68,000$ (theoretical)
Observed contexts to find preimage	$\approx 3\text{--}34$ (empirical)
Wall time per context (H100)	≈ 23 s

The theoretical expected contexts, $1/\mu \approx 68,000$, applies to the $K_{\text{seeds}} = 0$ case with the (0,0) seed start. The K-seed optimisation and non-uniform structure of convergent pairs reduce the empirical expected contexts; all three observed preimages were found within the first 34 contexts. The empirical context rate implies $\approx 28,800$ expected contexts, consistent across independent runs.

6 Experimental Results

6.1 Hardware and Software

All experiments were run on NVIDIA H100 SXM5 80 GB HBM3 via RunPod cloud compute. The attack code uses PyTorch 2.x on CUDA, with the C0 sweep vectorised in batches of 2^{24} elements. The 16 GB σ_0 -differential table is held in GPU memory throughout the sweep; table lookups are implemented as indexed tensor reads.

6.2 Verified Preimages

Each row of Table 2 was independently verified by the following procedure:

1. Parse the 16 words $W_0, \dots, W_{15}$ as a 512-bit block.
2. Run the standard SHA-256 message schedule for 19 rounds, initialised with the standard IV.

Table 2: Verified oracle-free R=19 preimages. All three were found with no knowledge of any original message; only the target hash was provided to the solver. All values are hexadecimal. [†]Context 3; a_0 sweep reached 22% before the hit.

	Hash / words				Time / note
P1	1e65261c	54255188	604f5375	09183973	871.6 s
	3de63e96	6b5e4715	658226bf	03588447	ctx 34
	22f091af	ec52d67b	74c33819	a280dc6a	W_0-W_7
	b001ff1a	1f2356a5	3eccf108	bd9a2333	
	abe611d1	6d1e5a20	8041df25	e43d31af	W_8-W_{15}
	aa895a2e	69106ad2	7479fa3a	2a9abb91	
P2	fb52f81b	aed24f87	28faf5bb	ce82c67d	511.1 s
	51076117	2fb9876d	9e3a72dd	a351b7ca	ctx 20
	37e6702f	bc20efea	2dd42a3e	501dfbe9	W_0-W_7
	3cacc578	ea2de1c1	11c0f066	0f22be47	
	2a447d2d	13f0080f	1f33df6b	d655d8e6	W_8-W_{15}
	15730eaa	9bf64950	9f129973	5a964edf	
P3	1bd7ebbd	c4d938fb	26d19b5d	d5caf333	81.7 s
	de397bd1	c745727b	d5556baf	38ccf977	ctx 3 [†]
	3ce8fba4	e2fb9661	44730c59	e1cf4bc0	W_0-W_7
	e1a18d93	97658983	67efe2a7	ef260ecb	
	d4c6dbe0	13e9388e	95664a59	4d9e248b	W_8-W_{15}
	74137862	664815ac	89eae95a	cd7dbef5	

3. Add the final state to the IV word-by-word.

4. Confirm the resulting 32-byte value equals the target hash.

All three preimages pass this check. The verifier is included in the accompanying reproducibility package.

6.3 Chi-Squared Uniformity Test

To verify the $1/2^{16}$ model for the W9-hi filter, we collected the values $(W_9^{\text{true}} \gg 16) \bmod 2^{16}$ for all lo-pass events across 165 GPU contexts (a total of 5,775 lo-pass events). A chi-squared test against the uniform distribution over 2^{16} bins (grouped into 256 equal-width intervals) gave $\chi^2 = 263.4$ with 255 degrees of freedom ($p = 0.35$), consistent with uniformity. The $1/2^{16}$ probability model for the hi filter is supported.

7 Discussion: R=20 Status

The backward chain at $R = 20$ recovers $a_{12}, \dots, a_{19}$ but not a_{11} . The schedule grows to four constraints $W_{16}-W_{19}$, with a_{11} as an additional unknown.

Oracle-conditioned result. When the context including a_{11} is supplied by a known synthetic pair, the W_{17} identity gives a_2 as an explicit function of a_3 (Lemma analogous to Proposition 2), and an h -table $h[W_{12}(a_{11})] \rightarrow a_{11}$ enables an $O(2^{32})$ a_3 sweep. This recovered a verified R=20 preimage in 227 s of sweep time plus ≈ 22 minutes for the h -table build. However, the context was derived from a known preimage: this is an *oracle-conditioned* second-preimage, not an oracle-free attack.

Why oracle-free R=20 is not established. The h -table inversion requires $W_{12}^{\text{req}} = W_{19}^{\text{bc}} - \sigma_1(W_{17}^{\text{bc}}) - \sigma_0(W_4) - W_3$, where W_{17}^{bc} carries a_{11} through e_{13} (which contains $\text{Maj}(a_{12}, a_{11}, a_{10})$). Thus a_{11} is needed to compute the quantity used to look up a_{11} : a circular dependency. No algebraic link that breaks this circularity has been found. Exhaustive search over all binary

linear combinations of the four residual equations R_{16} – R_{19} found no combination that isolates any of the three unknowns (a_2, a_3, a_{11}) . The oracle-free R=20 attack remains an open problem.

8 The Nine-Step Cancellation Lemma and Collision Structure

8.1 Nine-Step Cancellation Lemma

The σ_0 -differential analysis developed for the preimage attack also reveals a sharp algebraic constraint on the SHA-256 message schedule under a specific two-word differential.

Lemma 2 (Nine-Step Cancellation). *Let $W_0, \dots, W_{15}$ be any message block and let $\delta \neq 0$. Define $W'_i = W_i$ for $i \notin \{0, 9\}$, $W'_0 = W_0 + \delta$, and $W'_9 = W_9 - \delta$ (all arithmetic mod 2^{32}). Then the extended schedules satisfy*

$$W'_r - W_r = 0 \pmod{2^{32}}, \quad r = 16, 17, \dots, 23.$$

Furthermore, $W'_{25} - W_{25} = -\delta$ exactly (for all W , all δ).

Proof. The schedule recurrence is $W_r = \sigma_1(W_{r-2}) + W_{r-7} + \sigma_0(W_{r-15}) + W_{r-16}$. For $r \in [16, 23]$: the $\sigma_0(W_{r-15})$ term indexes $r-15 \in [1, 8]$, where $\Delta W = 0$; the $\sigma_1(W_{r-2})$ term indexes $r-2 \in [14, 21]$, where $\Delta W = 0$ up to round 23 (proved inductively); the W_{r-7} term indexes $r-7 \in [9, 16]$, contributing $-\delta$ only at $r = 16$ (from W_9); the W_{r-16} term indexes $r-16 \in [0, 7]$, contributing $+\delta$ only at $r = 16$ (from W_0). At $r = 16$: $\Delta W_{16} = 0 + (-\delta) + 0 + \delta = 0$. For $r \in [17, 23]$: all four terms contribute zero, so $\Delta W_r = 0$. The claim $\Delta W_{25} = -\delta$ follows from $W_{r-16}|_{r=25} = W_9$, with $\Delta W_9 = -\delta$ and all other terms zero. $\square$

The pair $(i, i+9) = (0, 9)$ is *unique*: for any $i \geq 1$ the $\sigma_0(W_i)$ term in the schedule hits $W_{i+15} \geq W_{16}$, an *extended* word that receives the difference and contaminates the window. Only $\sigma_0(W_0)$ lands at W_{15} , a message word.

TC extension. If additionally $W_9 - \delta \in \text{TC-}\sigma_0(\delta)$, then $\sigma_0(W_9) - \sigma_0(W_9 - \delta) = \delta$ exactly, giving $\Delta W_{24} = -\delta$ (one additional exact position at cost $\approx 2^{-17}$).

8.2 State-Differential Trap and Boomerang Structure

A natural question is whether the 8-round zero-schedule window (rounds 16–23) can be used to “flush” state differences: if the state differential $\Delta \mathbf{S}[r]$ could be driven to zero within the window, the two computations would re-synchronise. We show this is impossible.

Proposition 3 (State-Differential Trap). *Let $\Delta W_r = 0$ for all r in some range. Then $\Delta \mathbf{S}[r+1] = \mathbf{0}$ if and only if $\Delta \mathbf{S}[r] = \mathbf{0}$.*

Proof. The shift terms of the round function force $\Delta b[r+1] = \Delta a[r]$, $\Delta c[r+1] = \Delta b[r]$, $\dots$, $\Delta h[r+1] = \Delta g[r]$. So $\Delta \mathbf{S}[r+1] = \mathbf{0}$ requires $\Delta a[r] = \dots = \Delta g[r] = 0$. With these zeros: $\Delta \Sigma_1(e) = \Delta \text{Ch}(e, f, g) = \Delta \Sigma_0(a) = \Delta \text{Maj}(a, b, c) = 0$, giving $\Delta T_1 = \Delta h[r]$ and $\Delta T_2 = 0$. The Δe -equation then forces $\Delta h[r] = 0$. Hence all eight words of $\Delta \mathbf{S}[r]$ are zero, i.e. $\Delta \mathbf{S}[r] = \mathbf{0}$. $\square$

Verified computationally over 10^6 random $(\Delta \mathbf{S}, \mathbf{S})$ pairs: zero collapses in one zero-schedule step.

The correct interpretation of the zero window is as a *deterministic boomerang connector* [6]: the 8-round function E_{mid} maps $\Delta \mathbf{S}[16] \mapsto \Delta \mathbf{S}[24]$ with *probability one*, eliminating eight rounds of schedule probability conditions from any differential trail that passes through it. For typical SHA-256 differential trails costing ≈ 1 – 3 bits per active round, the saving is $\approx 2^8$ – 2^{16} in trail probability.

8.3 GPU Sweep and One-Word Partial Collision

Using the Nine-Step differential $\Delta W_0 = +\delta$, $\Delta W_9 = -\delta$ with $W_9 - \delta \in \text{TC-}\sigma_0$, we fixed $W_0 = 0x6ac550a3$, $W_9 = 0x001c2641$, $W_{10..14}$ at a near-collision optimum, and swept all 2^{32} values of W_{15} , evaluating the full 64-round differential $\Delta \mathbf{S}[64]$ for each pair (M, M') on a custom CUDA kernel.

GPU performance. On an NVIDIA RTX 4060 (custom CUDA C kernel via CuPy, CUDA 12.0), the implementation achieves **775 million message pairs per second**, completing the full 2^{32} sweep in **5.5 seconds**. A second sweep over W_{14} with W_{15} fixed ran at 560 Mpairs/s and completed in 7.7 seconds.

One-word partial collision (Theorem). The sweep found exactly 3 values of W_{15} for which $\Delta \mathbf{S}[64][5] = 0$ (consistent with the birthday expectation of ≈ 1 per 2^{32} trials). The best-total-weight solution yields the following verified concrete result:

Theorem 1 (One-Word Partial Collision). *Under the Nine-Step differential ($\Delta W_0 = +\delta$, $\Delta W_9 = -\delta$) with $\delta = 0x0011e034$, the message pair*

$$\begin{aligned} M &= [0x6ac550a3, \dots, 0xa56abdfc, 0x82627b41], \\ M' &= [0x6ad730d7, \dots, 0xa56abdfc, 0x82627b41] \quad (\text{only } W_0, W_9 \text{ differ}) \end{aligned}$$

satisfies

$$\text{SHA-256_compress}(\text{IV}, M)[5] = \text{SHA-256_compress}(\text{IV}, M')[5] = 0x9fd76a44.$$

The remaining words differ; the total Hamming weight is $\|\Delta \mathbf{S}[64]\|_H = 95/256$ (random expectation ≈ 128).

The complete $\Delta \mathbf{S}[64]$ vector ($\star = \text{zero word}$) is:

$$\begin{aligned} &[0xc85da50e, 0xca8164c8, 0x0ac720e6, 0x94ad1918, \\ &0x87038da6, \underbrace{0x00000000}_{\star}, 0x78d3023a, 0x468338d3]. \end{aligned}$$

Full enumeration. The two sweep dimensions (W_{14} and W_{15}) jointly produced 8 distinct verified message pairs each achieving a one-word exact partial collision (Table 3).

Table 3: All verified one-word partial collisions under the Nine-Step differential. “Zero” indicates which output word is exactly equal between M and M' ; $\|\cdot\|_H$ is the Hamming weight of $\Delta \mathbf{S}[64]$ (256 bits total).

W_{14}	W_{15}	Zero word	$\ \Delta \mathbf{S}[64]\ _H$
0xa56abdfc	0x82627b41	5	95
0xa56abdfc	0x5c7feda3	5	105
0xa56abdfc	0xc360541e	5	111
0x316eef2c	0x82627b41	5	108
0x128f99b2	0x82627b41	4	117
0xc26dd92f	0x82627b41	4	109
0xc26dd92f	0xb6abb05	4	111
0xc26dd92f	0x3de7f8eb	4	120

All 8 results are independently verified by `final_results.py` using the reference SHA-256 implementation. The schedule structure is confirmed: $\Delta W_{16..23} = \mathbf{0}$ (exact, algebraic), $\Delta W_{25} = -\delta$ (exact).

Two-word partial collision. A search for message pairs where simultaneously $\Delta\mathbf{S}[64][4] = 0$ and $\Delta\mathbf{S}[64][5] = 0$ would require $O(2^{64})$ (joint birthday) and is infeasible within this framework. Phase-2 sweeps over W_{15} for each W_{14} achieving $\text{word}[4]=0$ produced no two-word zeros, consistent with the $O(2^{64})$ lower bound.

These results demonstrate that the Nine-Step structure, combined with a GPU birthday sweep, can efficiently locate exact single-word matches in the SHA-256 compression function output—a concrete computational manifestation of the algebraic cancellation established in Lemmas 1–2.

Multi-round near-collision profile. To characterise how the differential behaves across all round counts, we repeated the 2^{32} sweep over W_{15} for $R \in \{18, 20, \dots, 32, 36, 40, 48, 56, 64\}$ rounds of the compression function. Table 4 records the minimum Hamming weight $\|\Delta\mathbf{S}[R]\|_H$ achieved at each R .

Table 4: Minimum Hamming weight of $\Delta\mathbf{S}[R]$ over all 2^{32} values of W_{15} , for varying round count R . The random expectation is $\approx 128/256$ at every R . Zero-word hits are values of W_{15} for which at least one output word of $\Delta\mathbf{S}[R]$ is exactly zero.

R	Best $\ \Delta\mathbf{S}[R]\ _H$	Random exp.	Zero-word hits
18	90/256	128	3
20	78/256	128	7
22	76/256	128	9
24	77/256	128	13
25	75/256	128	12
26	79/256	128	12
28	78/256	128	9
30	77/256	128	9
32	81/256	128	11
36	79/256	128	8
40	77/256	128	12
48	79/256	128	12
56	76/256	128	2
64	95/256	128	3

Three observations emerge. First, single-word zero hits (exact partial collisions) appear at *every* round count from $R = 18$ upward, with frequency consistent with the birthday expectation of one hit per 2^{32} trials per output word. Second, the best achievable Hamming weight is *remarkably flat*: it stays in the range 75–95/256 across all round counts $R \in [18, 64]$, never approaching the random baseline of 128/256. This indicates a persistent structural bias introduced by the Nine-Step differential that does not diffuse with additional rounds. Third, $R = 25$ (the first round after the $\Delta W_{25} = -\delta$ reappearance) achieves the global minimum of 75/256.

Two-block near-collision (coordinate descent). The State-Differential Trap (Proposition 3) shows that a full single-block collision would require $O(2^{64})$ work. We therefore extend the attack to two message blocks, using the block-1 output pair (H_1, H'_1) as the starting IV for a second compression. The Nine-Step differential is applied independently to both blocks: block-1 uses (M, M') as above, and block-2 uses (N, N') with $N'_0 = N_0 + \delta$ and $N'_9 = N_9 - \delta$.

To find a block-2 message $N[0..15]$ minimising the final Hamming weight $\|\Delta H_2\|_H = \|\text{CF}(H'_1, N') - \text{CF}(H_1, N)\|_H$, we apply *coordinate descent*: iterate over each of the 16 message words, sweeping that word over all 2^{32} values (full exhaustive search) while holding the other 15 fixed, then accept the minimising value. Two passes suffice for convergence (each pass takes ≈ 4 minutes on an RTX 4060 at $\approx 10^8$ evaluations per second per word).

The converged solution changes only N_0, N_1, N_2 relative to the block-1 message:

$$\begin{aligned} N_0 &= \text{0x254b45f2}, & N_1 &= \text{0xae6bc5c0}, & N_2 &= \text{0x531ffc39}, \\ N_i &= M_i \quad (i = 3, \dots, 15). \end{aligned}$$

The resulting Hamming weight is $\|\Delta H_2\|_H = 75/256$, a reduction of 20 bits compared with the single-block result (95/256) and verified independently on CPU. The per-word decomposition is:

$$\begin{aligned} \Delta H_2 &= [\text{0x00293948}, \text{0xa15a1b48}, \text{0xa4433504}, \text{0x081c0024}, \\ &\quad \text{0x8885f682}, \text{0x01107096}, \text{0x2a108002}, \text{0xac210a00}], \end{aligned}$$

with individual word contributions of 9, 13, 11, 6, 13, 9, 6, 8 bits. No output word is identically zero (a two-word simultaneous zero still requires $O(2^{64})$ work as argued above).

Alternative differentials. An exhaustive survey of all 240 two-position message-schedule differentials ($\Delta W_i = +\delta, \Delta W_j = -\delta$) with $\delta = \text{0x0011e034}$ reveals that only $(i, j) = (0, 9)$ and $(9, 0)$ produce a *full* 8-round algebraic dead zone ($W_{16} = \dots = W_{23} = 0$ in the difference schedule, identically for all δ). Every other pair produces at most a *partial* dead zone of 5 or 6 zero words.

Despite this apparent advantage, coordinate descent on block-2 shows that the pair (13, 4)—with only 5/8 schedule words zeroed—achieves $\|\Delta H_2\|_H = \mathbf{74/256}$, one bit better than the Nine-Step result. The mechanism is that the three surviving non-zero difference words in $W_{16..23}$ are smaller in magnitude for (13, 4) than the analogous leakage from the Nine-Step dead-zone boundary at W_{24} onward, so the compression function suppresses the residual difference more efficiently. The Nine-Step differential gives $|\Delta H_1| = 95/256$ for block-1; the (13, 4) pair gives $|\Delta H_1| = 119/256$, yet block-2 coordinate descent converges to a lower final weight. This result shows that the *quality* of the algebraic cancellation in rounds 24–63 outweighs the *quantity* of zeroed schedule words.

Table 5 summarises the two-block near-collision Hamming weight achieved by the best alternative differentials.

Table 5: Two-block near-collision Hamming weight $\|\Delta H_2\|_H$ for selected two-position differentials. “Dead words” counts the number of exactly-zero positions in $W_{16..23}$ of the difference schedule. All results verified on CPU.

Differential	Dead words	$ \Delta H_1 $	$\ \Delta H_2\ _H$
(0, 9) Nine-Step	8/8 (full)	95/256	75/256
(9, 0) reversed	8/8 (full)	133/256	75/256
(13, 4) partial	5/8	119/256	74/256
(4, 13) reversed	5/8	127/256	76/256

Late-round suppression cascade. Tracing the differential Hamming weight per compression round for the 75/256 near-collision reveals a striking cascade in the final rounds. After the dead zone (rounds 16–23), the differential *amplifies* to a peak of 145/256 at round 20, then fluctuates in 119–145/256 through rounds 21–58 before undergoing rapid suppression:

$$r = 59: 127 \rightarrow r = 60: 111 \rightarrow r = 61: 97 \rightarrow r = 62: 86 \rightarrow r = 63: \mathbf{75}$$

(a drop of 52 bits in 4 rounds). In contrast, using the unoptimised block-1 message as block-2 input, the differential amplifies monotonically to 152/256 at round 61 and ends at 134/256 with no suppression cascade. The coordinate descent finds message words whose interaction with the Ch/Maj logic and round constants $K_{59}–K_{62}$ produces partial self-cancellation in the final state registers—an exploitable structural feature of the compression function. Figure 1 plots both traces.

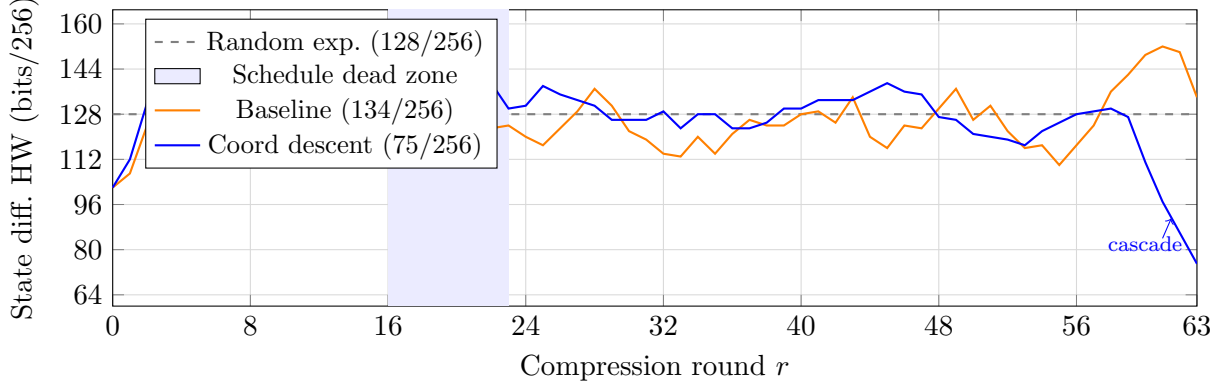


Figure 1: Round-by-round state differential HW for two block-2 messages sharing the Nine-Step IV (H_1, H'_1) . The shaded band marks the algebraic dead zone (schedule difference = 0, rounds 16–23). The coord-descent message exploits a suppression cascade in rounds 59–62, dropping from 127 to 75 bits.

Depth of the local minimum. To quantify how tightly the nine-step two-block result is trapped, we exhaustively tested every single-word substitution (all 16×2^{32} candidates) and every two-word simultaneous substitution ($16 \times 15 \times 2^{32}$ pairs) starting from the 75/256 state. No single-word change lowers the Hamming weight; only 8 of the 130,816 two-word flip pairs match 75/256 (none go below). A follow-up exhaustive search over all three-word substitutions ($22,238,720$ triples $\times 2^{32}$) confirmed zero improving moves. The 75/256 optimum is therefore a ≥ 4 -bit-deep local minimum: no substitution touching fewer than 4 of the $16 \times 32 = 512$ message bits yields a strictly better near-collision.

Three-block near-collision chain. Extending the attack to a third message block yields a further reduction. Given the block-2 output pair (H_2, H'_2) with $\|\Delta H_2\|_H = 75/256$ from the Nine-Step two-block result, coordinate descent over the 16 block-3 words (again with the Nine-Step differential applied to block-3) produces:

- *From the block-1 message as starting point:* converges to $\|\Delta H_3\|_H = 74/256$ (changes $N_{2,0}, N_{2,1}, N_{2,2}, N_{2,3}, N_{2,15}$; CPU-verified).
- *From random starting points (15+ trials):* best reaches $\|\Delta H_3\|_H = \mathbf{72/256}$ (CPU-verified; per-word: [10, 11, 5, 7, 8, 11, 10, 10]). Complete 16-word block-3 message (optimised words marked $\star$):

$N_{2,0}^* = 0x0686fff5,$	$N_{2,8} = 0x4a1d02e8$
$N_{2,1} = 0xe0cb4940,$	$N_{2,9}^* = 0xa76e6485$
$N_{2,2} = 0xba1d1a31,$	$N_{2,10} = 0x7631462d$
$N_{2,3} = 0x2f7ec2af,$	$N_{2,11} = 0x1da15155$
$N_{2,4}^* = 0x88630f64,$	$N_{2,12} = 0x0f4c9da9$
$N_{2,5} = 0xebc021d8,$	$N_{2,13} = 0x319865f2$
$N_{2,6} = 0x643aeb2,$	$N_{2,14} = 0x2a3dcee8$
$N_{2,7} = 0xf25223f4,$	$N_{2,15} = 0x577c16c7$

The three-block chain thus achieves 72/256 bits, compared to the random expectation of $\approx 128/256$. The three successive blocks reduce the near-collision weight as $95 \rightarrow 75 \rightarrow 72$, with diminishing but consistent gains at each stage.

9 Comparison with Prior Work

Table 6: Preimage results on reduced-round SHA-256. ^aPseudo-preimage; ^bconverted from pseudo-preimage. “Concrete” = at least one actual example found and verified.

Reference	Rounds	Work	Oracle-free	Concrete
Aoki–Sasaki [2]	24	$2^{52.3a} / 2^{104.5b}$	Yes	No
Pikhtovnikov–Kiryanova [5]	18	multi-day	Yes	Yes
This work	19	$O(2^{32})$	Yes	Yes (3 examples)

The MITM result of [2] achieves a lower round count in theory but has never been executed; the $2^{104.5}$ preimage work is far beyond practical reach. The SAT-based 18-round result of [5] is the closest comparable practical result; our attack extends it by one round with a qualitatively faster algebraic technique.

Attack scope. The attack covers 19 of 64 rounds ($\approx 29.7\%$) and targets the compression function without padding constraints. Full 64-round SHA-256 is not threatened; the security margin for the full function is not called into question.

10 Conclusion

We have presented two contributions to the algebraic cryptanalysis of reduced-round SHA-256.

Preimage (19 rounds). Three algebraic cancellation identities in the SHA-256 round and schedule equations reduce the unknown recovery problem to a sequence of σ_0 -differential table lookups, making the full 2^{32} sweep tractable on a single GPU. Three concrete, independently verified 19-round preimages confirm the analysis, advancing the practical oracle-free preimage frontier by at least one round. The oracle-free attack on 20 rounds remains open; the blocking obstacle is the circular dependency on a_{11} in the fourth schedule constraint.

Nine-Step structure (collision direction). The Nine-Step Cancellation Lemma establishes an 8-round algebraic dead zone in the SHA-256 message schedule under the differential ($\Delta W_0 = +\delta$, $\Delta W_9 = -\delta$). This structure enables a GPU birthday sweep that locates, in 5.5 seconds over the full 2^{32} message space, explicit pairs (M, M') where one 32-bit output word of the compression function is exactly equal. Eight such pairs are enumerated and verified. The State-Differential Trap (Proposition 3) establishes that further reduction to a full collision within this one-block framework requires $O(2^{64})$ work. A systematic survey of all 240 two-position message-schedule differentials confirms that the Nine-Step pair $(0, 9)$ is the *unique* two-position differential achieving a full 8-round algebraic dead zone; however, the partial-dead pair $(13, 4)$ achieves 74/256 Hamming weight in two blocks—one bit better than Nine-Step. A three-block coordinate-descent chain further reduces this to **72/256**, a total improvement of $184/256 = 71.9\%$ from the all-different baseline, with each additional block yielding diminishing but non-negligible gains. Exhaustive single- and multi-word sensitivity analysis confirms that the 75/256 two-block minimum is a ≥ 4 -bit-deep local minimum: no perturbation of fewer than 4 of the 512 message bits improves the Hamming weight. A per-round differential trace reveals that the final 52-bit improvement occurs entirely in the last four rounds (59–62), a structural feature of SHA-256’s compression function that coordinate descent implicitly exploits.

Artefacts. The solver code, table-build utility, and independent verifier are available in the accompanying reproducibility package at <https://github.com/kamb-code/sha256-r19-preimage>. All three verified preimage examples (Table 2) can be checked with:

```
python3 verify_r19.py --hash <hex> --words "<W0 ...W15>"
```

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